

Bitwise differential trail through **WARP** found by SAT

CiVerLy

August 19, 2026

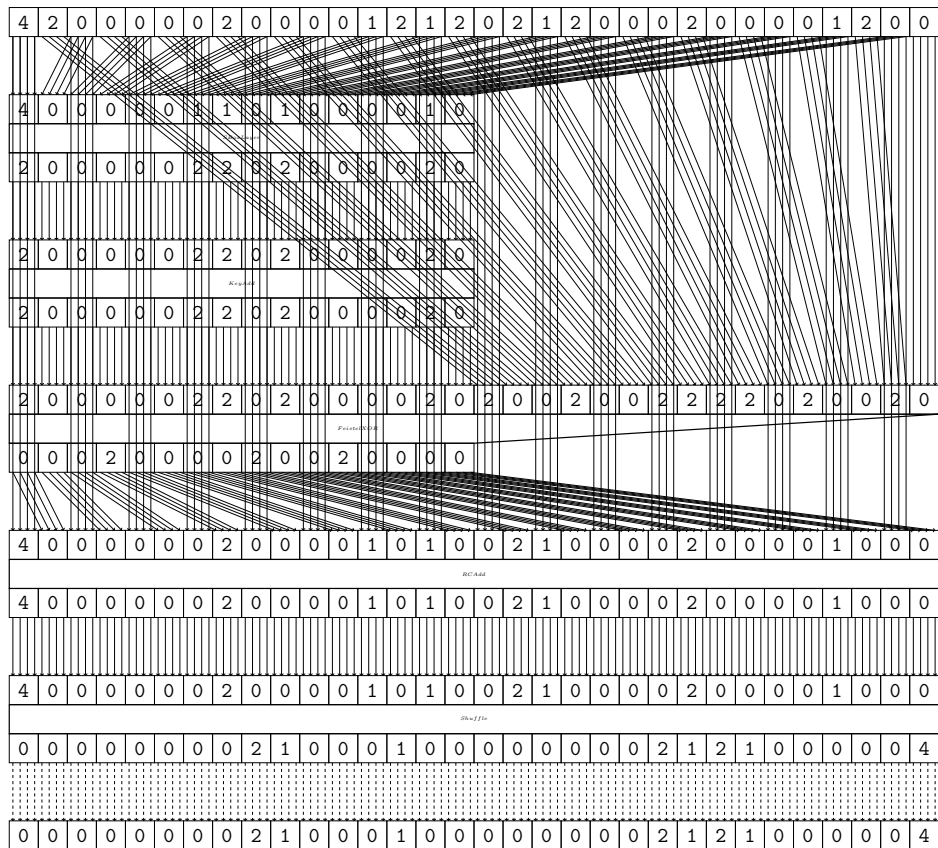
1 WARP

Maximal differential probability: 2^{-44}

4	2	0	0	0	0	0	2	0	0	0	0	1	2	1	2	0	2	1	2	0	0	0	2	0	0	0	0	1	2	0	0	
4	2	0	0	0	0	0	2	0	0	0	0	1	2	1	2	0	2	1	2	0	0	0	2	0	0	0	0	1	2	0	0	
unpermut																																
0	0	0	0	0	0	0	2	1	0	0	0	1	0	0	0	0	0	0	0	0	0	2	1	2	1	0	0	0	0	4		
unpermut																																
0	0	0	0	0	0	0	2	1	0	0	0	1	0	0	0	0	0	0	0	0	0	2	1	2	1	0	0	0	0	4		
unpermut																																
0	0	0	0	1	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	2	0	0	0	
unpermut																																
0	0	0	0	1	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	2	0	0	0	
unpermut																																
2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	2	
unpermut																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	2	
unpermut																																
0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	
unpermut																																
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	
unpermut																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
unpermut																																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
unpermut																																
0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	4	0	0	0	0	2	0	2	4	0	0	0	0	0	0	0	0	0
unpermut																																
0	0	0	0	0	0	0	0	0	2	0	0	0	0	4	0	0	0	0	2	0	2	4	0	0	0	0	0	0	0	0	0	0
unpermut																																
1	0	0	2	0	2	0	0	0	0	4	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0
unpermut																																
1	0	0	2	0	2	0	0	0	0	4	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0
unpermut																																
2	0	0	4	0	0	2	0	0	4	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	1
unpermut																																
2	0	0	4	0	0	2	0	0	4	0	0	2	0	2	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0

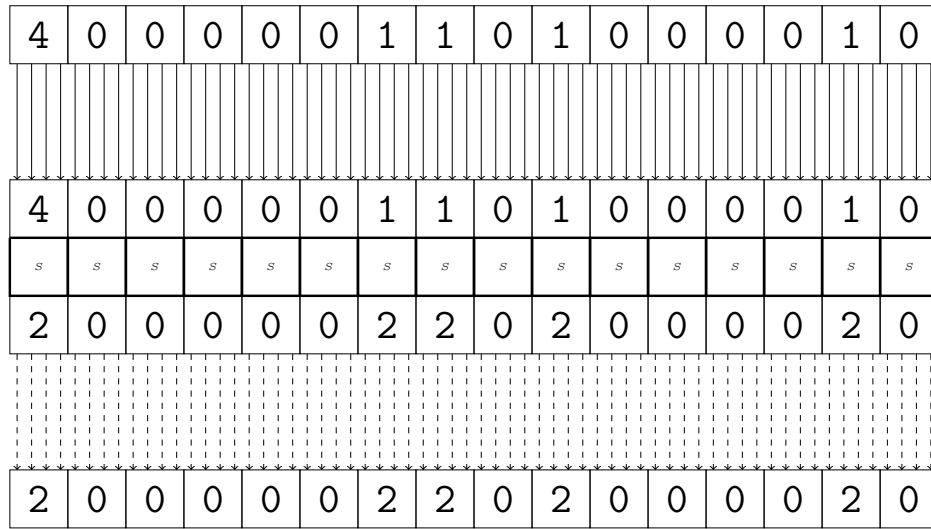
2 warp_round

Maximal differential probability: $2^{-10.0}$



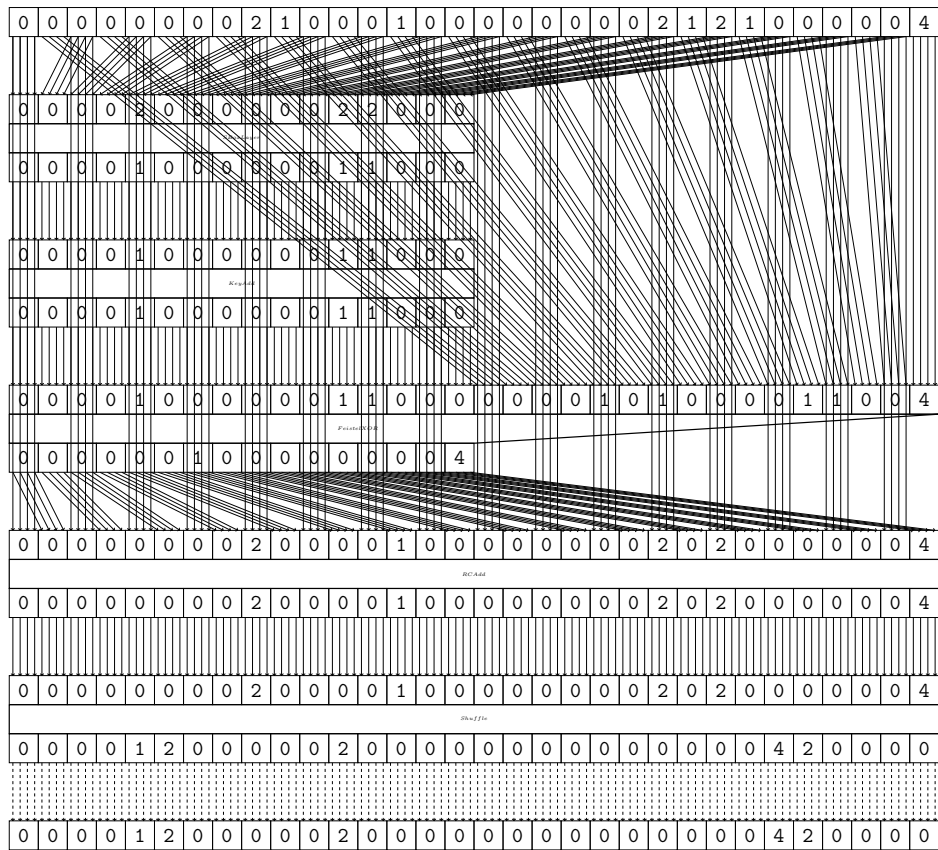
3 SBoxLayer

Maximal differential probability: $2^{-10.0}$



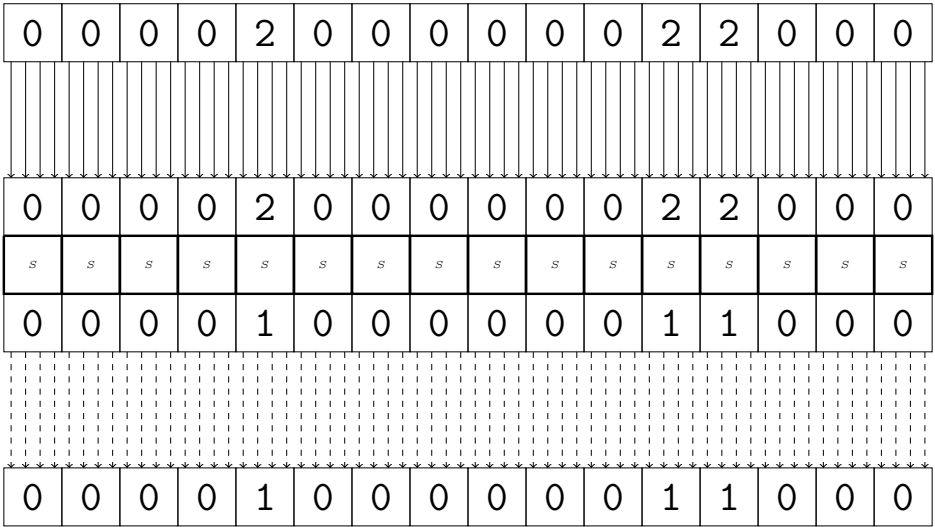
4 warp_round

Maximal differential probability: $2^{-6.0}$



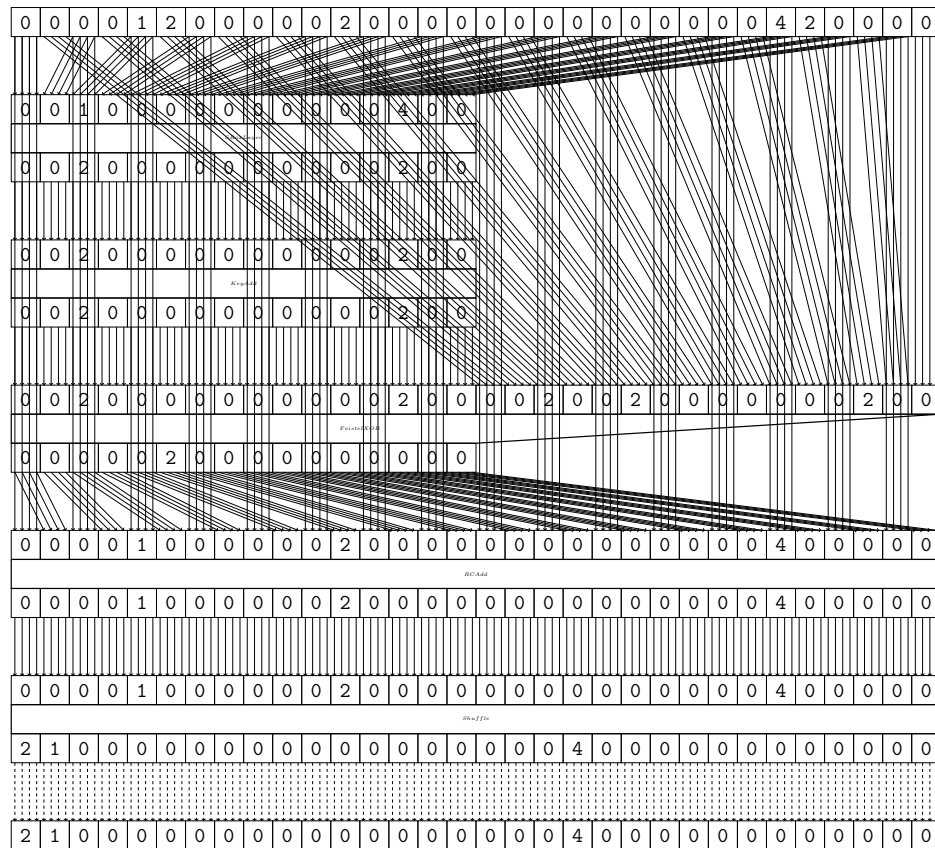
5 SBoxLayer

Maximal differential probability: $2^{-6.0}$



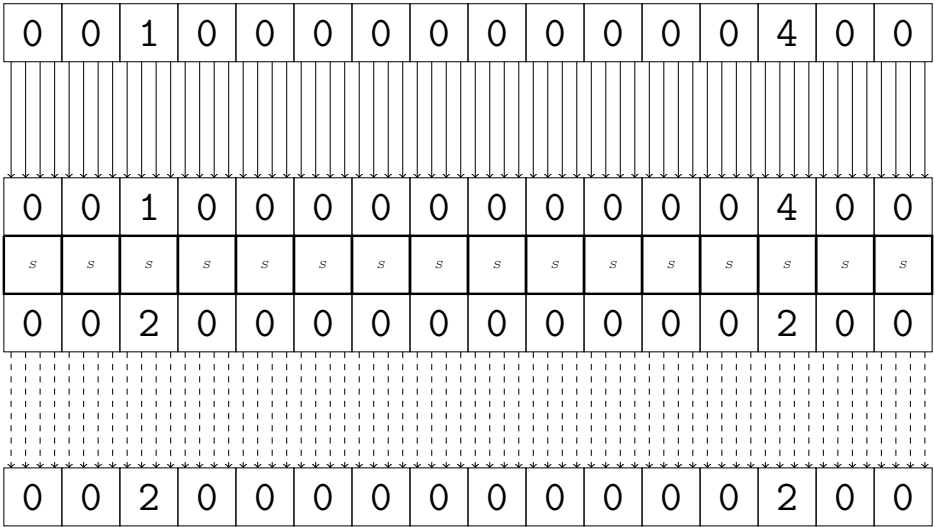
6 warp_round

Maximal differential probability: $2^{-4.0}$



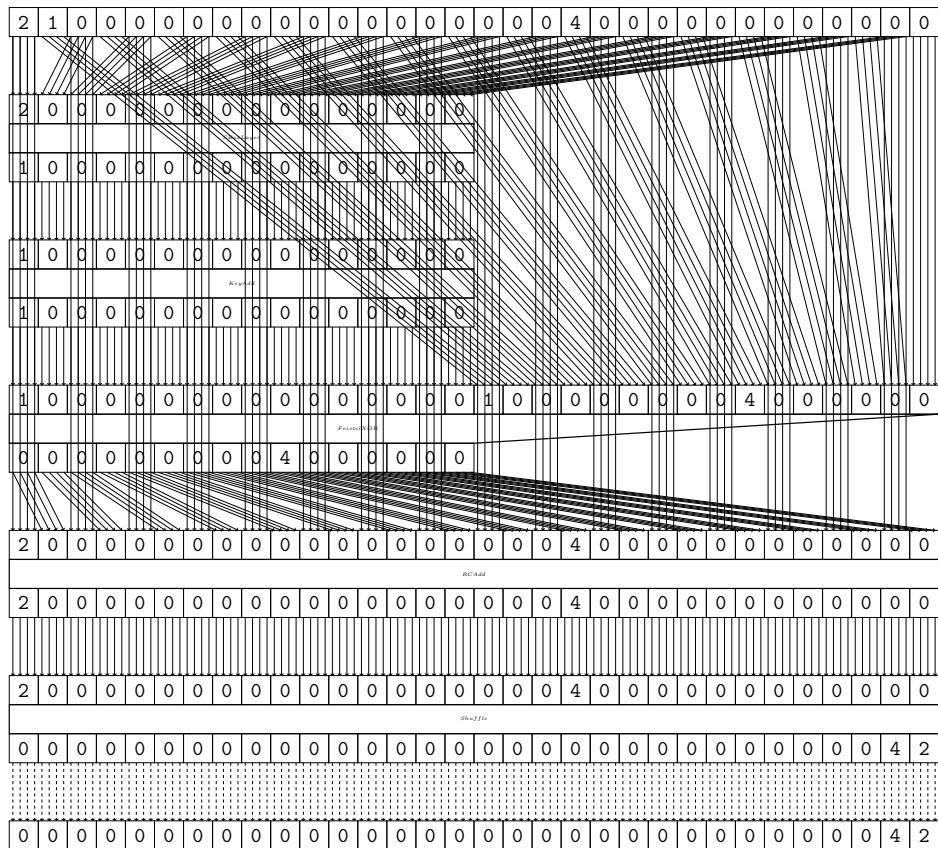
7 SBoxLayer

Maximal differential probability: $2^{-4.0}$



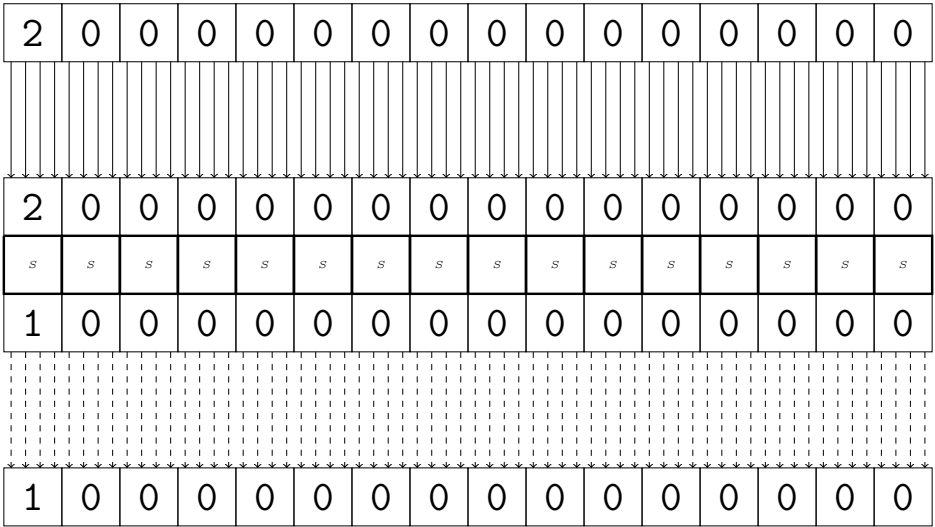
8 warp_round

Maximal differential probability: $2^{-2.0}$



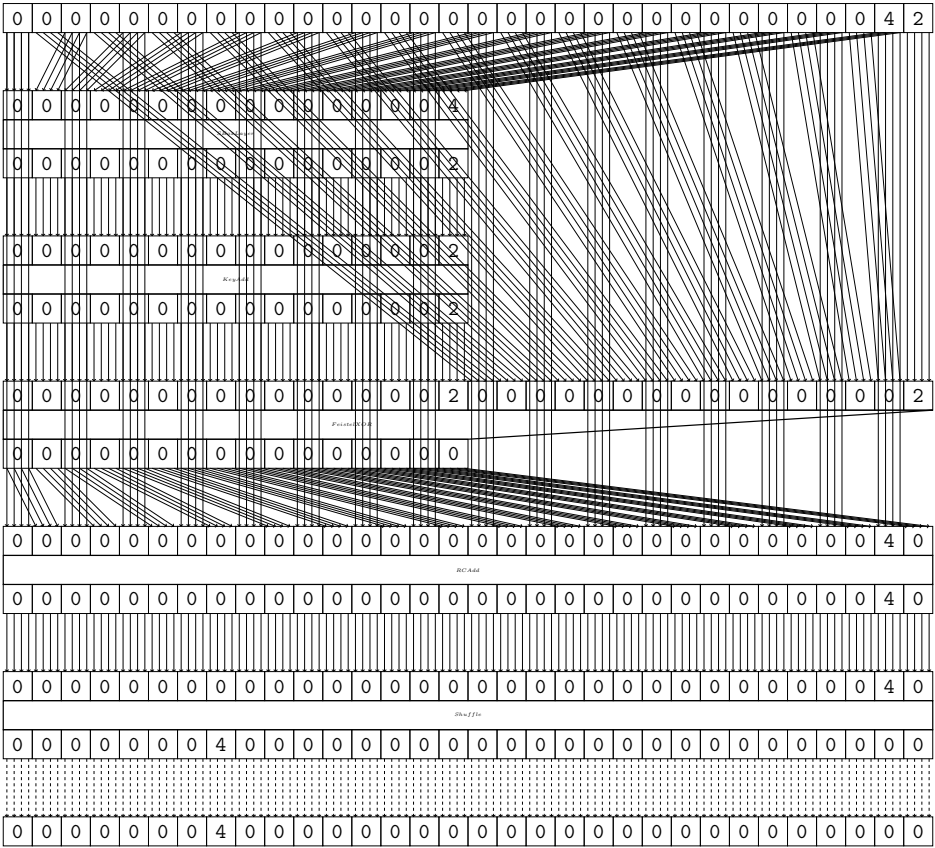
9 SBoxLayer

Maximal differential probability: $2^{-2.0}$



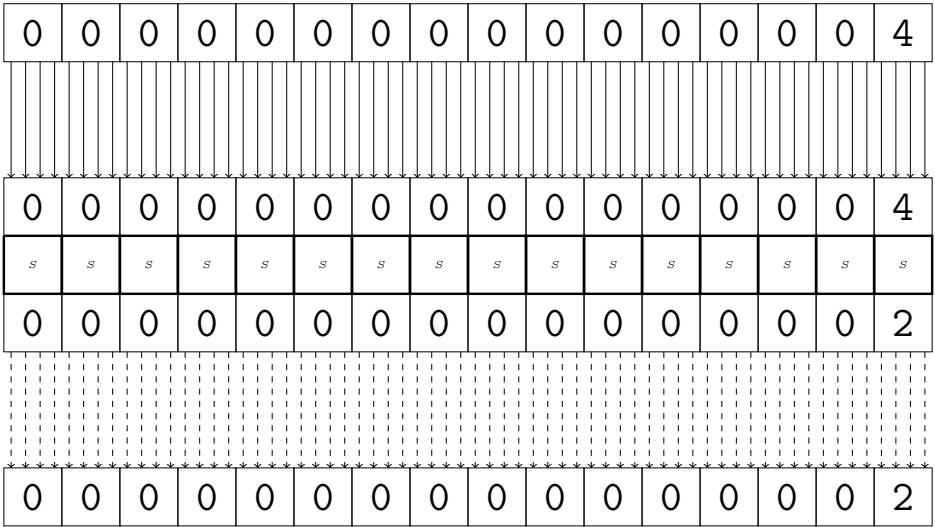
10 warp_round

Maximal differential probability: $2^{-2.0}$



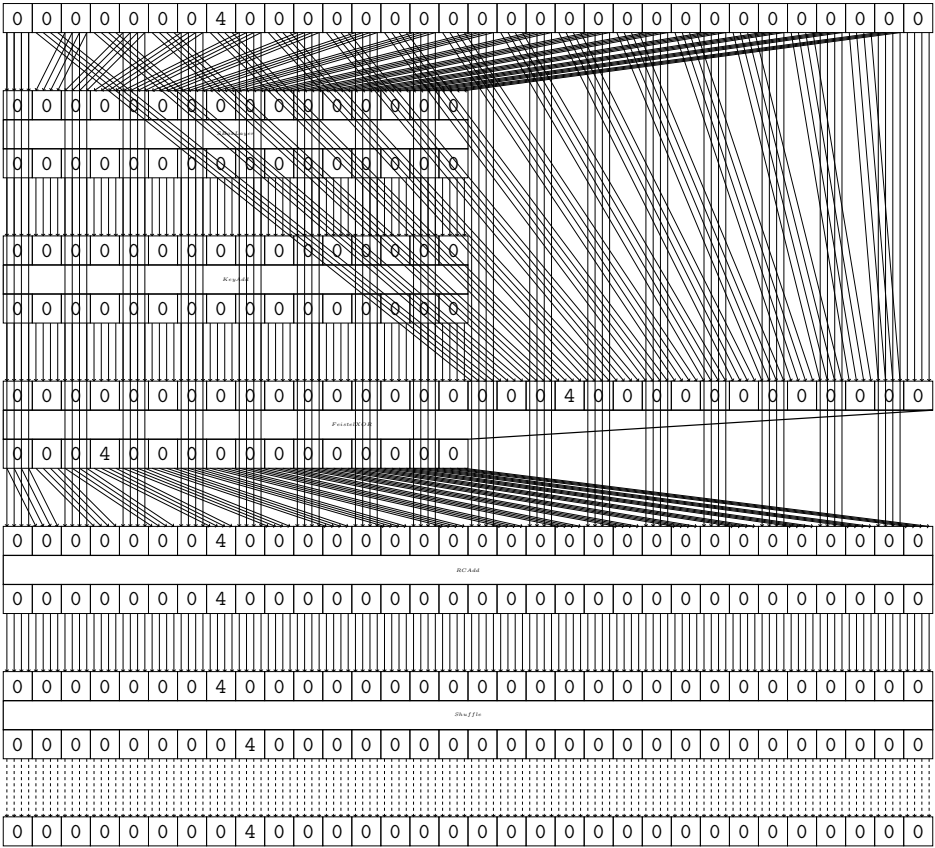
11 SBoxLayer

Maximal differential probability: $2^{-2.0}$



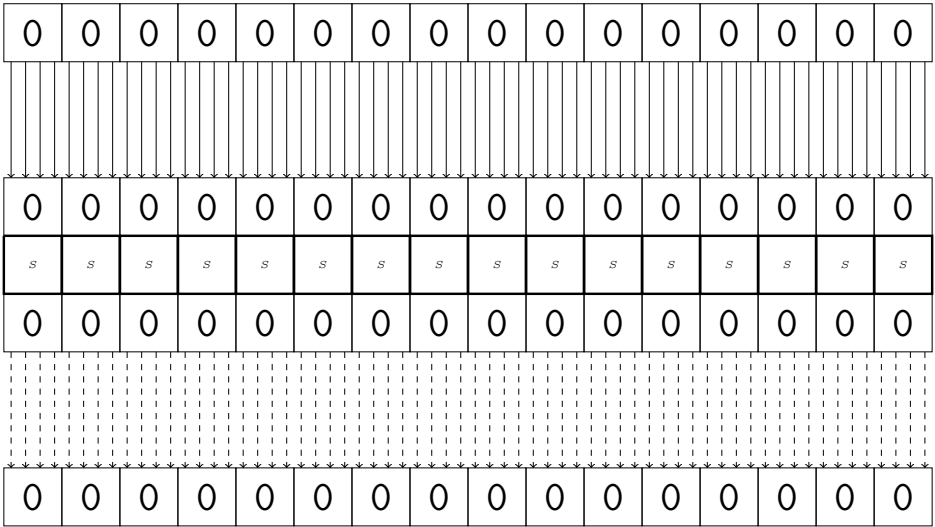
12 warp_round

Maximal differential probability: $2^{-0.0}$



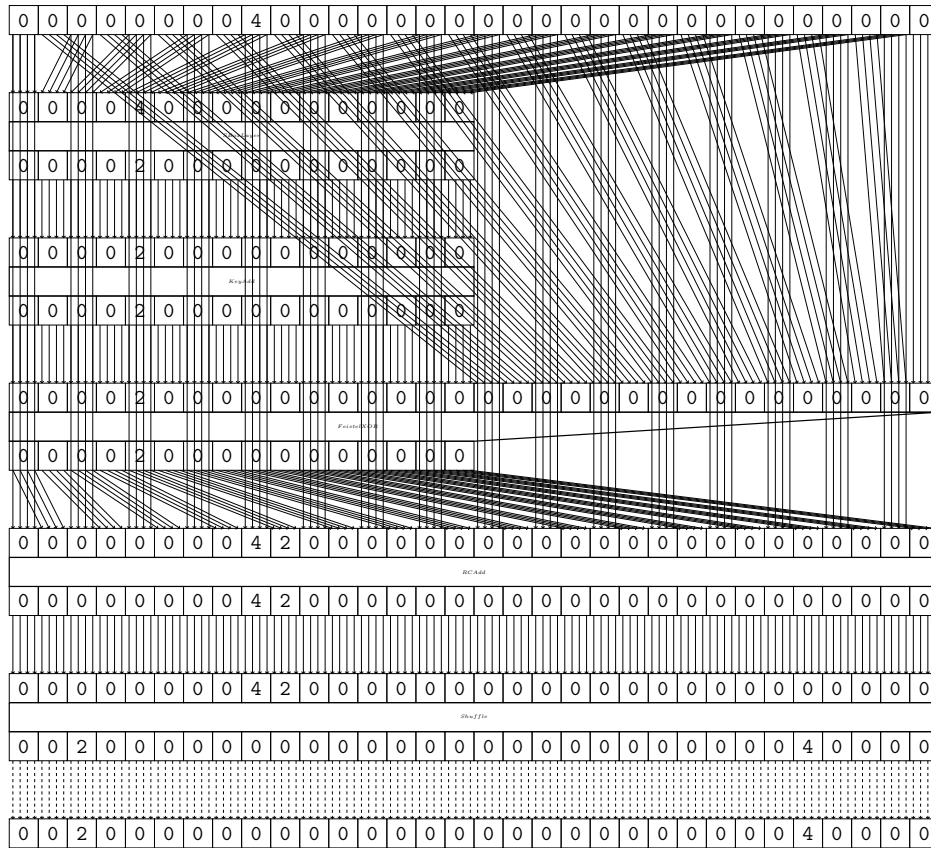
13 SBoxLayer

Maximal differential probability: $2^{-0.0}$



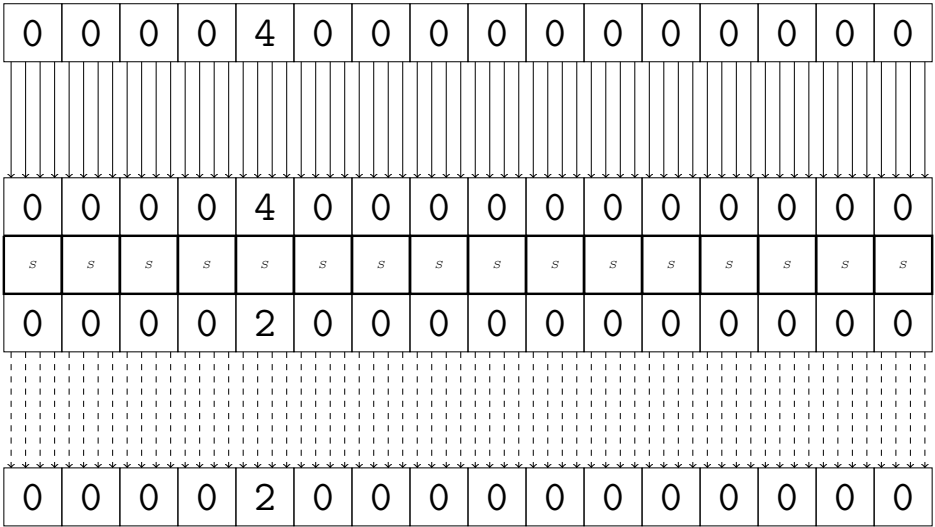
14 warp_round

Maximal differential probability: $2^{-2.0}$



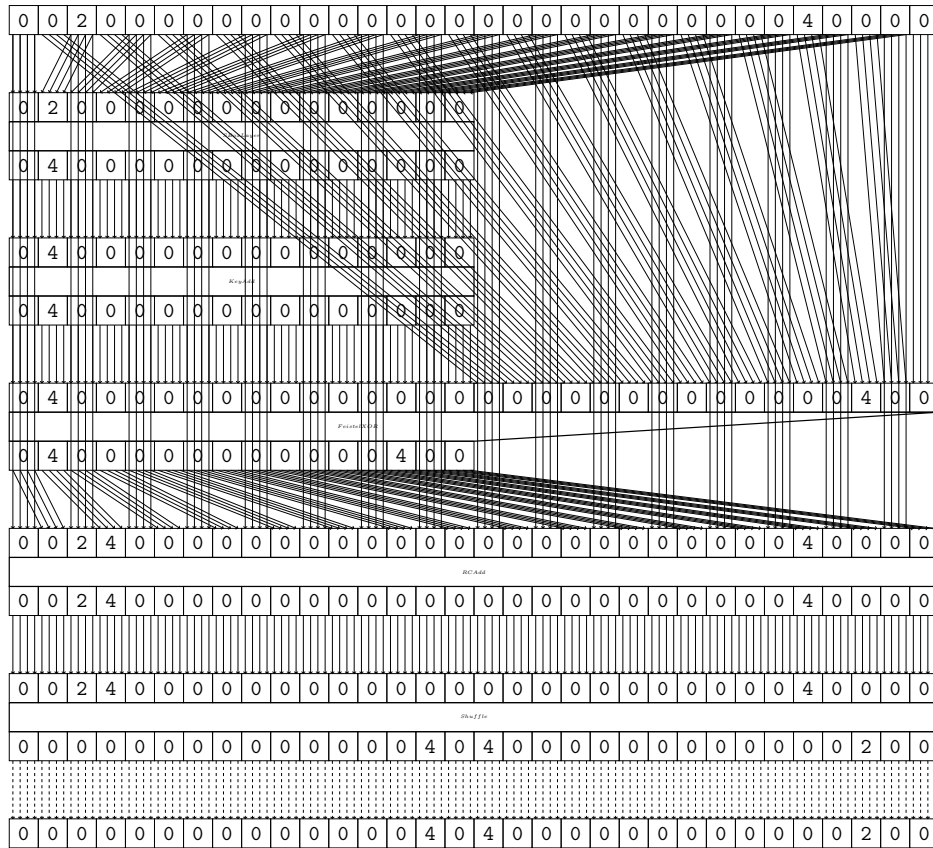
15 SBoxLayer

Maximal differential probability: $2^{-2.0}$



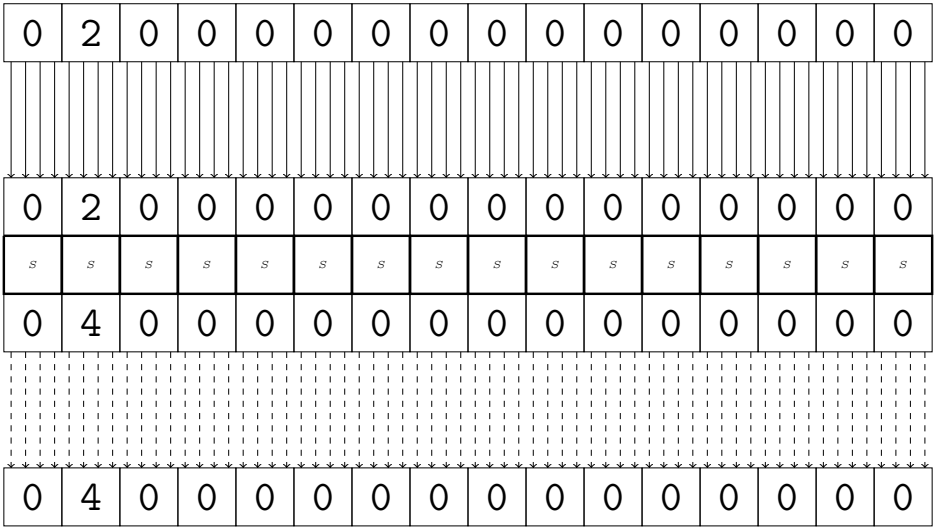
16 warp_round

Maximal differential probability: $2^{-2.0}$



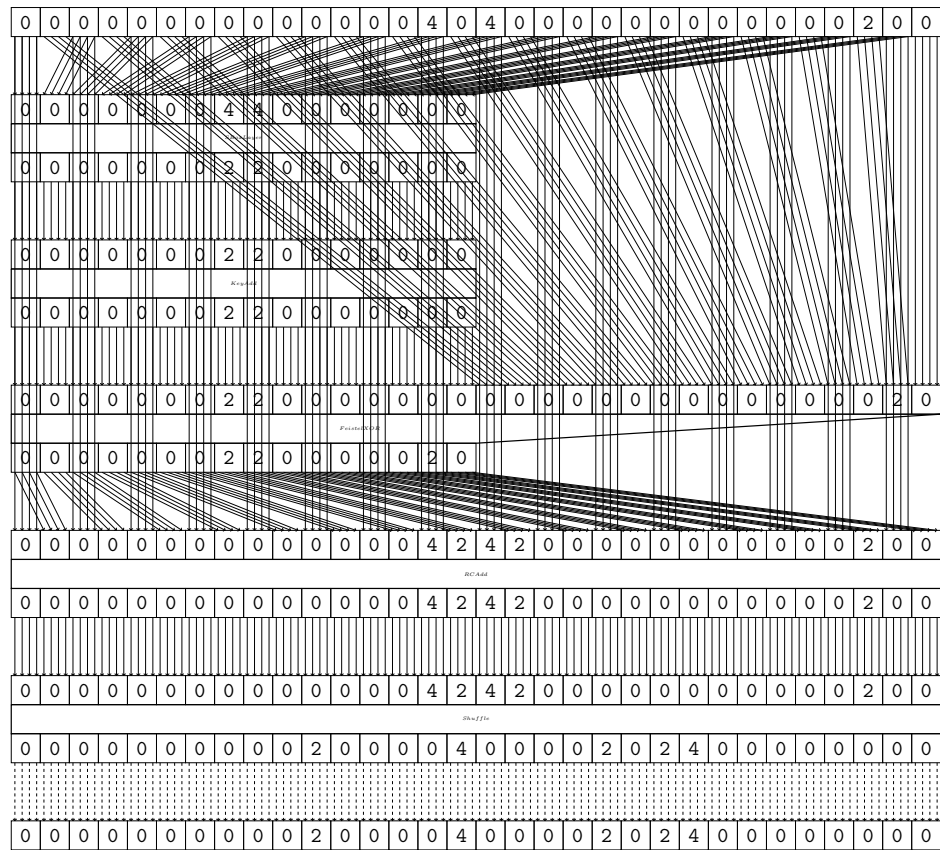
17 SBoxLayer

Maximal differential probability: $2^{-2.0}$



18 warp_round

Maximal differential probability: $2^{-4.0}$



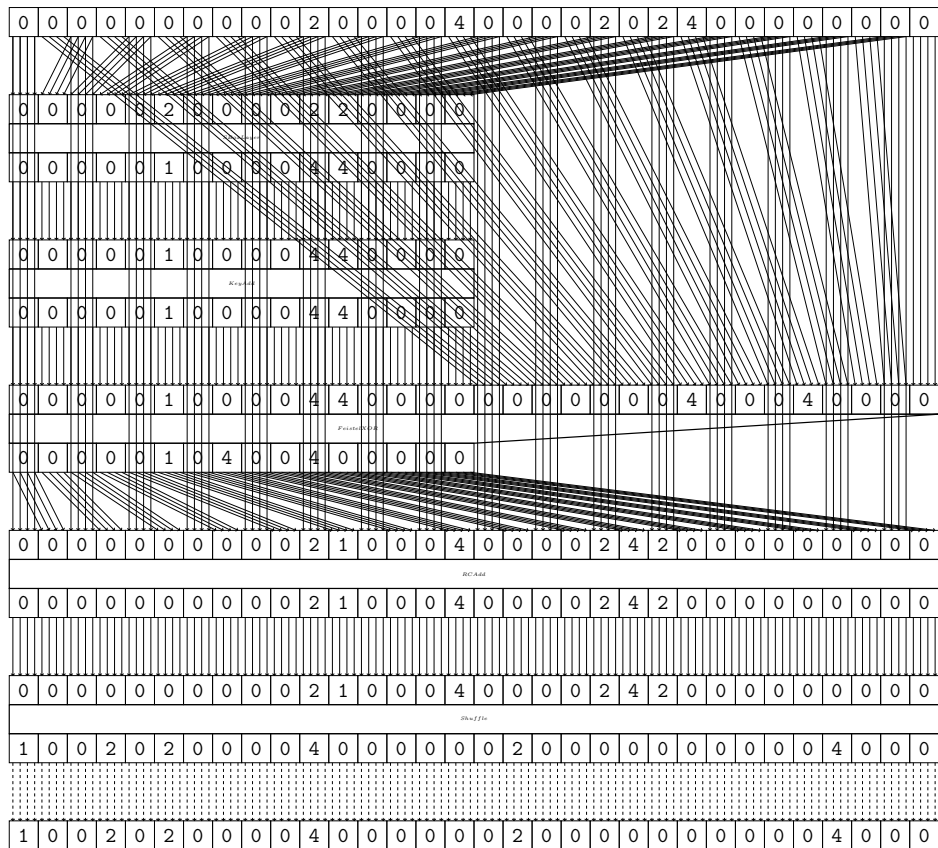
19 SBoxLayer

Maximal differential probability: $2^{-4.0}$

0	0	0	0	0	0	0	4	4	0	0	0	0	0	0	0
0	0	0	0	0	0	0	4	4	0	0	0	0	0	0	0
<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>
0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0
0	0	0	0	0	0	0	2	2	0	0	0	0	0	0	0

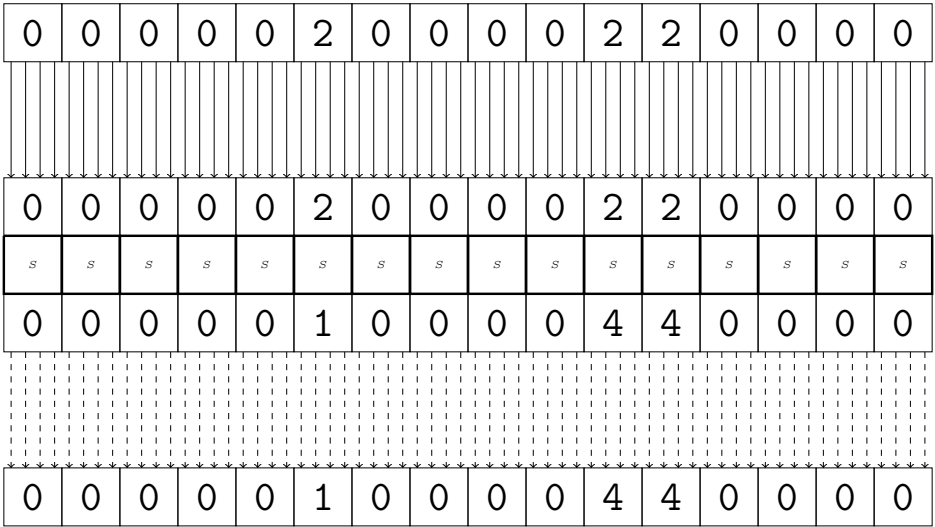
20 warp_round

Maximal differential probability: $2^{-6.0}$



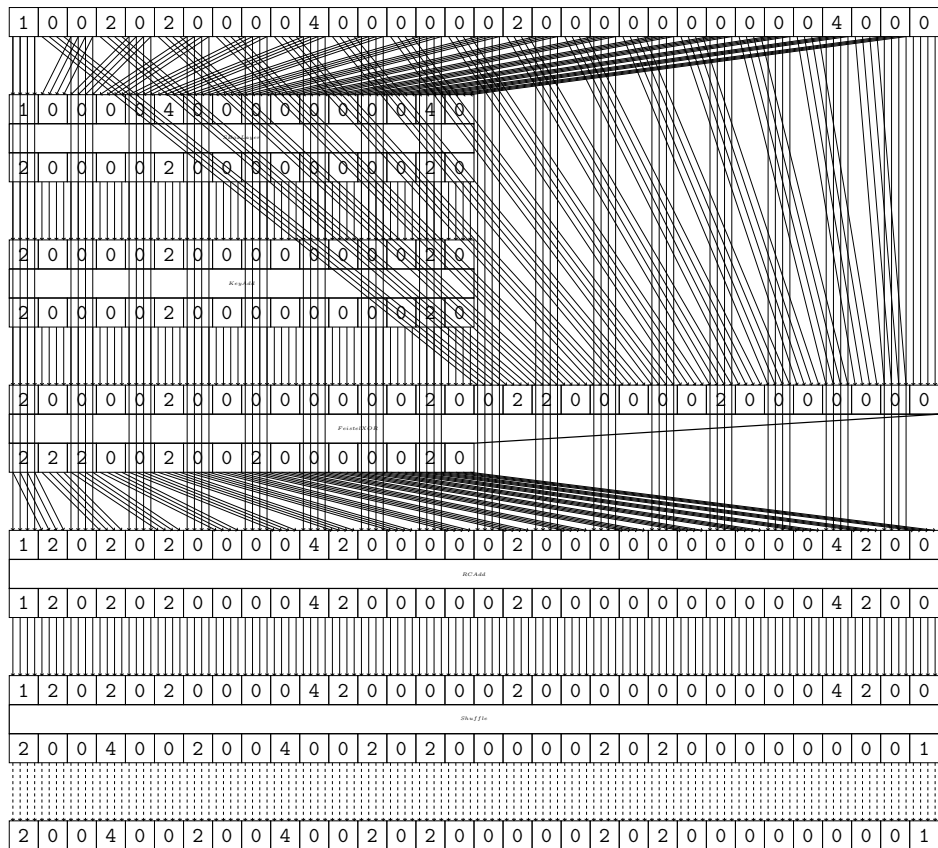
21 SBoxLayer

Maximal differential probability: $2^{-6.0}$



22 warp_round

Maximal differential probability: $2^{-6.0}$



23 SBoxLayer

Maximal differential probability: $2^{-6.0}$

